

Strings and the Wave Equation

One of the real beauties of physics is that systems that are governed by the same equation or set of equations will behave in fundamentally the same way. There's a huge variety of instruments that produce a huge variety of sounds, but down deep they're described by the same equation¹. We'll look at how a string behaves when it's under tension since that is the prototypical example of the sorts of behavior musical instruments show.

1. Forces in a String

We'll start by assuming we have a mass that has strings tied to either side of it, and those strings are under the same tension T . A string can only exert a force along the string, and we call that force the tension. If I want to pull something to the left using a string, I have to have the string pulling to the left. If the tensions in the pair of strings are equal but they're in opposite directions, then there's no net force. If you look at Figure 1, you'll see that even though there are two strings pulling at an angle, because they pull in opposite directions, the mass won't move.

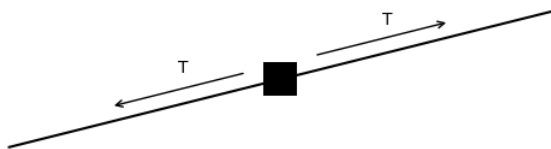


Fig. 1.— Mass under string tension. Because the strings are on opposite sides, there is no *net* force on the mass. If there's no force, then there's no acceleration, and the mass will happily sit in the same spot for all eternity.

The story is entirely different if the strings don't pull in exactly opposite directions. Picture a mass flat on a table with strings pulling in opposite directions. Now you lift the mass up a little bit. The strings are *mostly* pulling in opposite directions, but there is now a component of the string tension that pulls downwards. This component is in the *same* direction for both strings, so

¹Strictly speaking, this is only *mostly* true. Pianos are a familiar example, and the slight corrections to the governing equation directly explains why they're so hard to tune.

the vertical force doesn't cancel. This shouldn't come as a surprise - if you say pluck a stretched rubber band, the rubber band wants to go back to being straight. If you let it go, the rubber band will snap back to its original, straight shape.

Quantitatively, if the string has tension T , and the strings make an angle θ with the ground, then the horizontal force from each string is $T \cos(\theta)$, but in *opposite* directions so there is no net horizontal force. The vertical force is $T \sin(\theta)$ but now in the same direction, so the total force is $2T \sin(\theta)$. If we don't pluck the string very far, then θ will be a small angle, and the sine of a small angle is very close to the angle, $\sin(\theta) \sim \theta$. That means that the downwards force is close to $2T\theta$. To give you a sense for how good this approximation is, let's say we pluck a violin string by 1 cm. The string is about 60 cm long, so if we pluck in the middle, then we have a right triangle that is 30 cm along one side and 1 cm along the other. That means our angle is about $1/30$ radians, and if we take $\sin(1/30)$ we find that it is just 0.02% smaller than $1/30$. Assuming the angles are small is going to be a *very* good approximation.

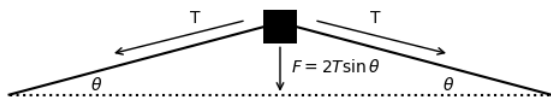


Fig. 2.— Mass under string tension. Now the strings are not pulling in opposite directions. Most of the force cancels out, but the small fraction pointing down points in the same direction for both strings. The mass will feel a downward force. It will start to move downwards, just like a plucked string.

2. String Motion

I'm now going to try to show you how we might expect a plucked string to behave. The math to actually do this right is well beyond the scope of this class, but I'll do my best to give you an idea of where the solution comes from. If it doesn't make sense, that's OK - you aren't responsible for any of the derivation. You will be responsible for being able to use the solutions, though, so you can skip to the end of this section if you like.

If we're going to look at an actual string, we're going to have to make our picture look like Figure 2 somehow. In a real string, we don't have a bunch of little masses with short strings connecting them. We don't have any masses at all, in fact, but the string itself has some mass. We can, however, draw a box around a little bit of a string. Then the mass we need to worry about is the mass of the string inside the box. The force on that bit of string will come from the direction of the strings just on either side of the box. We'll now assume we have a box of width dx and try to work out the forces and accelerations involved.

2.1. Force on a String Segment

If the string just has a constant slope then there won't be any net force because the forces will cancel. However, if the slope of the string is changing, then we will get a force because now the strings on either side of our box aren't pulling in the same direction. We'll still assume the string is almost horizontal, so the net force is still going to be up or down. We'll also assume that the slope of the slope is constant. How do we work out the force? Let's start by stepping back a level. If I told you a line has a slope m , then the change in my height if I move a distance dx will just be mdx . That's exactly what we mean by a slope - height over distance. Now if my slope has a slope c , and I want to know how much my slope changed over a distance dx , then the change in slope (*not* the change in height) is cdx .

If you look at Figure 2 carefully, you'll see that since each string makes an angle θ with the horizontal, the angle between the two strings is 2θ . That gives rise to a force of $2T\theta$. If the angle between the two strings was only θ , then the force would be $T\theta$. If we go back to the single string picture, then the angle between the two strings will be the amount the slope changes over our box, or cdx , which is just the angle between the two strings. The force on our box is then $Tcdx$ if our box is dx wide.

2.2. String Acceleration

Now that we have a force, we probably want to figure out an acceleration using Newton's second law, $F = ma$, or $a = F/m$. That means we need to figure out m , the mass of the string inside our box that is dx long. If the total string is x metres long, then the fraction of the string inside our box is dx/x . If the whole string has mass m , then the fraction of the mass inside our box is dx/x , and the total mass inside the box is mdx/x , which we'll write as $dx \frac{m}{x}$. The ratio $\frac{m}{x}$ is the mass of the string divided by its length, or the mass per unit length which we'll call λ . In our homework example, violin strings had masses of around 1 gram for a length of 60 cm, so $\lambda = 0.001\text{kg}/0.6\text{m} = 6 \times 10^{-4}\text{kg/m}$. As you can see, the units of λ , are kg/m . We can swap in λ to get a neater expression for the mass of the string inside our box: λdx .

Once we have the mass, we can work out the acceleration of the piece of string inside our box. Since $a = F/m$ and $F = Tcdx$ and $m = \lambda dx$, we have:

$$a = \frac{cTdx}{\lambda dx} = \frac{cT}{\lambda} \quad (1)$$

If you think about it for a minute, I hope you can see we should have expected the size of the box would go away. If I look at a piece of the string, very close-by points should have the same acceleration. If I draw my box around points with similar acceleration, it can't matter how big the box is because everyone has the same acceleration. The scaling of things makes sense as well - if the tension increases, our acceleration increases. If the mass increases, then the acceleration decreases. If we pull harder and bend the string more (increasing c), the string pulls back harder and the acceleration increases.

Since the acceleration is the rate of change of velocity, and velocity is the rate of change of position, the left-hand side is the rate of change of the rate of change of the (vertical) position of the string. On the right-hand side, the tension and the mass-per-length are constant, but c is the slope of the slope of the string position. If you'll indulge me in a bit of calculus notation, we can write down the fundamental equation for a vibrating string:

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\lambda} \frac{\partial^2 y}{\partial x^2} \quad (2)$$

The $\frac{\partial}{\partial}$ notation means that we're looking at the change of the thing in the top as we change the thing in the bottom. When we square it, that means we take not the change of the top, but the change of the change of the top. In words, the equation is telling is that the change of the change in string height with time equals the change of the change of string height with position, scaled by the tension divided by the mass-per-length. This equation is known as a *partial differential equation*, and you emphatically are not responsible for how to solve one. However, you might guess that because this equation is so simple (the slope of the slope in time is proportional to the slope of the slope in position) that it shows up frequently in physics. It does, and is called *the wave equation*. It governs waves on a string, sound in the air, and even light moving in space. Because these phenomena are governed by the same equation, they all have the same behavior. *If you've seen one wave, you've seen them all.*

3. Solutions of the Wave Equation

I'm not going to go through how to solve the wave equation from scratch, but I do want to show you what the solutions look like. I'll also show you that once we guess the correct solution, we can indeed *check* that it's correct. Unlike equations you may have seen before, partial differential equations (PDEs) generally are solved by arbitrary functions with some constraint on the arguments. In our case solutions look like $y = f(kx - \omega t)$, where f can be *any* function.

To check this works, we do need to figure out one fact about slopes. Let's say we're driving up a mountain, and it takes 20 km to climb the 1 km mountain. The average slope of our drive is then $1/20 = 0.05$. However, if we shortened the road to 10 km, but still had to climb the same 1 km, our new slope is $1/10 = 0.1$. If I have to climb the same height but change the distance over which I do it by a factor a , then my new slope increases by a . If $f(x)$ goes from 0 to 1 when x goes from 0 to 20, our function has a slope of 0.05, just like the mountain. Now what happens to our slope if we have $f(2x)$? Well, we know originally that $f(20) = 1$ and because we didn't change f , that still has to be true. However if I'm taking $f(2x)$, then when $x = 10$, I'm taking

$f(2 \times 10) = f(20) = 1$. I only changed x by 10, but still had to go up my 1 km, so the slope of f is now 0.1. More generally, if the slope of $f(x) = f'(x)$ where f' is some new function, then the slope of $f(ax) = af'(ax)$ for the same reasons that climbing the hill picked up a factor of 2. We don't actually have to worry about how to calculate f' - that's a real calculus problem, but it turns out we just need to know that the slope of $f(ax)$ is $af'(ax)$. If we take the slope of the slope (denoted f''), we pick up another factor of a , so the slope of the slope of $f(ax) = a^2 f''(ax)$, where $f''(ax)$ is the slope of the slope if I ignored the a .

The other thing we need to know is that if I'm looking for the slope along the x -direction, I don't need to worry about what happens with time. Similarly, if I'm looking for the slope along the time direction, I don't need to know what happens in the x -direction. We can now plug in $f(kx - \omega t)$ into the wave equation and see what happens. We know

$$\frac{\partial^2 f}{\partial t^2} = \omega^2 f''(kx - \omega t) \quad (3)$$

Similarly,

$$\frac{\partial^2 f}{\partial x^2} = k^2 f''(kx - \omega t) \quad (4)$$

We don't know how to calculate f'' , but that's OK - all we actually need is to know about those factors of ω^2 and k^2 , and that f'' is the same as we change time or position. If we plug these expressions into our equation, we have

$$\omega^2 f''(kx - \omega t) = k^2 \frac{T}{\lambda} f''(kx - \omega t) \quad (5)$$

We can cancel the f'' from both sides of the equation, leaving us with

$$\omega^2 = k^2 \frac{T}{\lambda} \quad (6)$$

That is, if I say tell you a value for k , then as long as I pick a value for ω that satisfies Equation 6, then my function $f(kx - \omega t)$ is a solution to the wave equation. As we saw in the propagating signals note, this is an arbitrary shape (we can pick whatever we want for f) that is moving along the string with velocity $\sqrt{\frac{\omega}{k}}$. If you've been paying close attention, you might have noticed that we have an equation for ω^2 . Since the square of a negative number is positive, we could just as well make ω negative and it will still solve the wave equation. That means our full solution is actually $f(kx - \omega t) + g(kx + \omega t)$. We can have anything for f and g , but we do know the speeds at which the shapes move. We have f moving left, and g moving right along the string. As we saw earlier $\frac{\omega}{k}$ is the speed at which a signal moves, so waves must move along our string with speed $\sqrt{\frac{T}{\lambda}}$ in both the left and right directions.